



9. FEATURE MATCHING UNDER ILLUMINATION CHANGES



Computer Vision – Image Matching



(a) CONCEPT EXPLANATION

Q. (a) Analyze why matching fails and explain how preprocessing or robust descriptors improve performance.

When the same scene is captured under different lighting conditions (e.g., bright vs. dark, shadows, or sunlight vs. overcast), the pixel intensities change significantly. This affects feature detection and description, causing:

- Keypoints may be missed in dark regions or overexposed areas.
- Descriptors become inconsistent for the same physical points.
- More mismatches and fewer correct correspondences.
- Lower matching accuracy.

Preprocessing (e.g., histogram equalization, gamma correction, Retinex) normalizes illumination and enhances visibility.

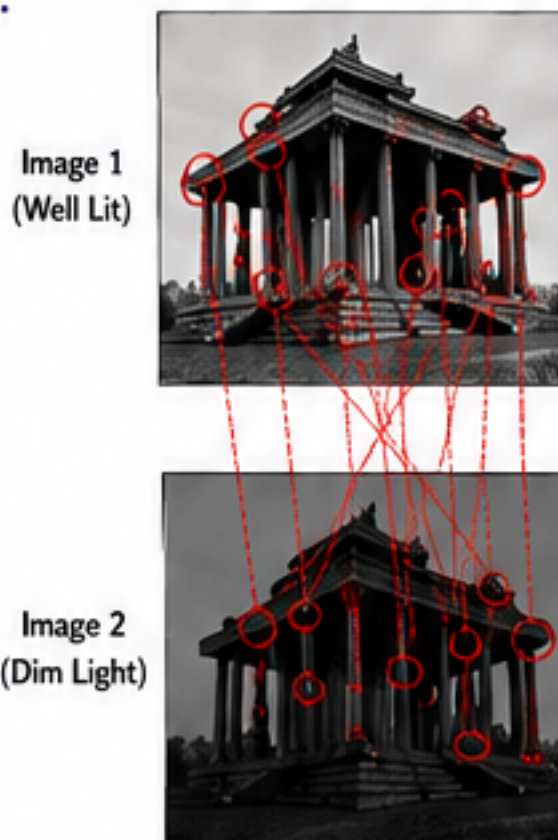
Robust descriptors (e.g., SIFT, SURF, ORB) are designed to be invariant or less sensitive to illumination changes.

Why Robust Descriptors Help

- Use gradients or intensity patterns that are less affected by global illumination changes.
- Capture local structures instead of raw intensities.
- Provide more consistent features across varying lighting.

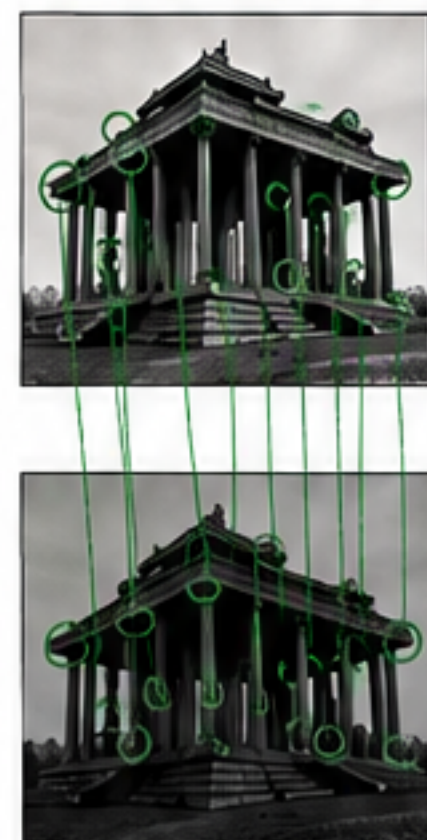
Effect of Illumination Changes on Feature Matching

Traditional Matching (Fails)



Many Incorrect Matches

Robust Matching (Succeeds)

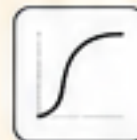


Many Correct Matches

Common Preprocessing Techniques



Histogram Equalization



Gamma Correction



Retinex



Contrast Enhancement

Benefits

- ✓ Better feature visibility
- ✓ More stable keypoints
- ✓ Fewer mismatches (outliers)
- ✓ Higher matching accuracy

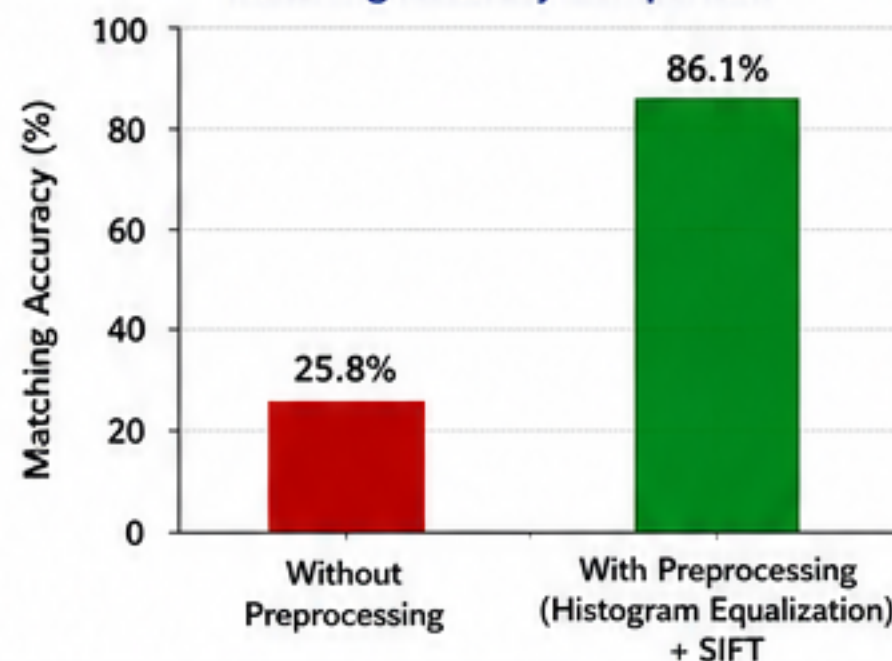


(b) NUMERICAL EXAMPLE

Q. (b) Consider two images of the same scene under different illumination. Compare feature matching performance before and after preprocessing (using histogram equalization) with SIFT descriptors.

Method	Detected Keypoints (No.)	Correct Matches (No.)	Incorrect Matches (No.)	Matching Accuracy (%)
Without Preprocessing (Traditional Matching)	680	82	236	25.8%
With Preprocessing (Histogram Equalization) + SIFT Descriptors	620	298	48	86.1%

Matching Accuracy Comparison



What does the result mean?

- Without preprocessing, illumination changes reduce correct matches and accuracy.
- Histogram equalization improves visibility and consistency.
- SIFT descriptors further enhance robustness to lighting variations.
- This leads to many more correct matches and higher accuracy.

Conclusion:

Illumination changes cause matching failures due to intensity variations. Preprocessing normalizes lighting, and robust descriptors (like SIFT) provide illumination-invariant features, significantly **improving matching performance**.



(c) VISUALIZATION

Input Images (Different Illumination)



Image 1 (Bright)

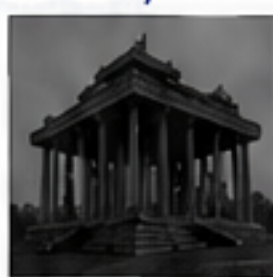
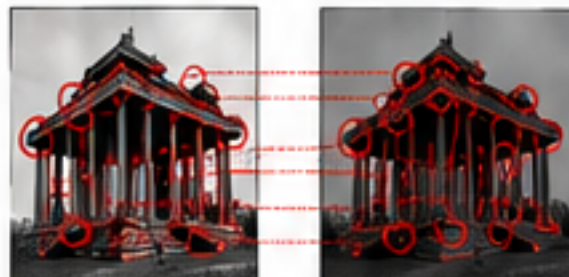


Image 2 (Dim)

Traditional Matching (Without Preprocessing)



Few / Incorrect Matches

After Preprocessing (Histogram Equalization)



Robust Matching with SIFT (After Preprocessing)



Many / Correct Matches



Observation: Illumination variations degrade feature matching. Preprocessing (e.g., histogram equalization) improves image quality, and robust descriptors (e.g., SIFT, SURF, ORB) ensure reliable matching.



Computer Vision



Image Matching



Robust Descriptors



Illumination Handling



Better Accuracy